Alimentazioni e masse (3V3, 5V_SYS, GND_A, GND_D, V1P35, V1P4, VBUS1, VIN_SYS) sono etichette globali nei fogli: non sono disegnate qui.

CPU

- DDR_A[0..15]
- DDR_BA[0..2]
- DDR_DQ[0..15]
- DDR_CAS_N
- DDR_CKE
- DDR_CK_N
- DDR_CK_P
- DDR_CS_N
- DDR_LDM
- DDR_LDQS_N
- DDR_LDQS_P
- DDR_ODT
- DDR_RAS_N
- DDR_RESET_N
- DDR_UDM
- DDR_UDQS_N
- DDR_UDQS_P
- DDR_VREF
- DDR_WE_N
- SOC_DDR_ZQ
- DAC61402_FAULT
- ECSPI1_SCLK
- ECSPI2_SCLK
- MAIN_HIT_N
- MAIN_I2C_SCL
- MAIN_I2C_SDA
- MAIN_SPI_ADDR_0
- MAIN_SPI_ADDR_1
- MAIN_SPI_ADDR_2
- MAIN_SPI_ALERT_N
- MAIN_SPI_CS_MASTER
- MAIN_SPI_MISO
- MAIN_SPI_MOSI
- MAIN_SPI_PREFIX_0
- MAIN_SPI_PREFIX_1
- MAIN_SPI_PREFIX_2
- PWR_SPI_DAC_CS
- PWR_SPI_INA239A_CS
- PWR_SPI_INA239B_CS
- PWR_SPI_MISO
- PWR_SPI_MOSI
- VBIAS_ENABLE
- VREF_ENABLE
- I2C1_SCL
- I2C1_SDA
- PHY_MDC
- PHY_MDIO
- PHY_RST_N
- RMII_CRSDV
- RMII_REFCLK
- RMII_RXD0
- RMII_RXD1
- RMII_RXER
- RMII_TXD0
- RMII_TXD1
- RMII_TXEN
- BT_DIS_N
- BT_HOST_WAKE
- BT_UART_CTS
- BT_UART_RX
- BT_UART_TX
- BT_WAKE_HOST
- SD2_CLK_S
- SD2_CMD
- SD2_D0
- SD2_D1
- SD2_D2
- SD2_D3
- WL_DIS_N
- WL_WAKE_HOST
- GNSS_EXTINT
- GNSS_PPS
- GNSS_UART_RX
- GNSS_UART_TX
- USB1_DM
- USB1_DP
- USB2_DM
- USB2_DP
- USB2_OC_N
- USB2_VBUS_EN
- QSPI_CS1_N
- QSPI_CS_N
- QSPI_D0
- QSPI_D1
- QSPI_D2
- QSPI_D3
- QSPI_SCLK_S
- SD1_CD_N
- SD1_CLK_S
- SD1_CMD
- SD1_D0
- SD1_D1
- SD1_D2
- SD1_D3
- POR_B
- PWR_SRC_ST

File: cpu.kicad_sch

DDR3L

- DDR_A[0..15]
- DDR_BA[0..2]
- DDR_DQ[0..15]
- DDR_CAS_N
- DDR_CKE
- DDR_CK_N
- DDR_CK_P
- DDR_CS_N
- DDR_LDM
- DDR_LDQS_N
- DDR_LDQS_P
- DDR_ODT
- DDR_RAS_N
- DDR_RESET_N
- DDR_UDM
- DDR_UDQS_N
- DDR_UDQS_P
- DDR_VREF
- DDR_WE_N
- SOC_DDR_ZQ

File: ddr3l.kicad_sch

Bus MIKILAB

- DAC61402_FAULT
- ECSPI1_SCLK
- ECSPI2_SCLK
- MAIN_HIT_N
- MAIN_I2C_SCL
- MAIN_I2C_SDA
- MAIN_SPI_ADDR_0
- MAIN_SPI_ADDR_1
- MAIN_SPI_ADDR_2
- MAIN_SPI_ALERT_N
- MAIN_SPI_CS_MASTER
- MAIN_SPI_MISO
- MAIN_SPI_MOSI
- MAIN_SPI_PREFIX_0
- MAIN_SPI_PREFIX_1
- MAIN_SPI_PREFIX_2
- PWR_SPI_DAC_CS
- PWR_SPI_INA239A_CS
- PWR_SPI_INA239B_CS
- PWR_SPI_MISO
- PWR_SPI_MOSI
- VBIAS_ENABLE
- VREF_ENABLE

File: bus-mikilab.kicad_sch

Rete

- I2C1_SCL
- I2C1_SDA
- PHY_MDC
- PHY_MDIO
- PHY_RST_N
- RMII_CRSDV
- RMII_REFCLK
- RMII_RXD0
- RMII_RXD1
- RMII_RXER
- RMII_TXD0
- RMII_TXD1
- RMII_TXEN

File: rete.kicad_sch

Radio

- BT_DIS_N
- BT_HOST_WAKE
- BT_UART_CTS
- BT_UART_RX
- BT_UART_TX
- BT_WAKE_HOST
- SD2_CLK_S
- SD2_CMD
- SD2_D0
- SD2_D1
- SD2_D2
- SD2_D3
- WL_DIS_N
- WL_WAKE_HOST

File: radio.kicad_sch

GNSS

- GNSS_EXTINT
- GNSS_PPS
- GNSS_UART_RX
- GNSS_UART_TX

File: gnss.kicad_sch

USB

- USB1_DM
- USB1_DP
- USB2_DM
- USB2_DP
- USB2_OC_N
- USB2_VBUS_EN

File: usb.kicad_sch

Storage

- QSPI_CS1_N
- QSPI_CS_N
- QSPI_D0
- QSPI_D1
- QSPI_D2
- QSPI_D3
- QSPI_SCLK_S
- SD1_CD_N
- SD1_CLK_S
- SD1_CMD
- SD1_D0
- SD1_D1
- SD1_D2
- SD1_D3

File: storage.kicad_sch

Alimentazione

- POR_B
- PWR_SRC_ST

File: alimentazione.kicad_sch

Sheet: /
File: SchedaIMX6ULL-SD.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/10

U1 NXP MCIMX6Y2CVM08AB (LWX 6UL) full feature, 792 MHz, -40..105 C. MAPBGA-289 14x14 mm passo 0.8). Stevo: IMX6ULLCEC Rev 1.2 tabelle 90/91, confrontate con la netlist di una scheda KiCad aperta: 289/289.

Boot: BOOT_MODE[1:0]=10 (boot interno) + BOOT_CFG tutti bassi con i pull-down interni da 100k (tabella 78) = QSPI, come SW501 tutto OFF sulla EVK NXP. Pull-up 2NP su LCD_DATA04 e LCD_DATA06 (micro50). Se QSPI e' vuota il ROM va in download seriale su USB_OTG1.

Assegnazione: ECSPI2=BOOT_POWER, ECSPI1=MAIN_I2C2=bus MIKROAB, I2C1=EEPROM MAC, UART1=console, UART3=GNSS, UART8=BT, GPT1 CAPTURE1=PPS (UART2_TX), CAPTURE2=EXTINT (UART2_RX). Elenco completo in docs/pinmap-imx6ull.md.

U1A
MCIMX6Y2CVM08AB
DDR1

U1B
MCIMX6Y2CVM08AB
Alimentazioni

U1C
MCIMX6Y2CVM08AB
Masse

U1D
MCIMX6Y2CVM08AB
I2C / BOOT_CFG

U1E
MCIMX6Y2CVM08AB
NAND QSPI uSDHC2 GPIO1 e SD1

U1F
MCIMX6Y2CVM08AB
I2C ECSPI1,2 UART3 e ENET2

U1G
MCIMX6Y2CVM08AB
ENET1 UART GPIO1 JTAG

U1H
MCIMX6Y2CVM08AB
USB_clock SNVS sistema

J5
Console UART1

Pull-up open drain USB2_OC_N PWR_SRC_ST

#FLG150 PWR_FLAG
#FLG151 PWR_FLAG
#FLG152 PWR_FLAG
#FLG153 PWR_FLAG
#FLG154 PWR_FLAG
#FLG155 PWR_FLAG

VDD_ARM_CAP / VDD_SOC_CAP: uscite LDO interni, solo condensatori

VDD_HIGH_CAP (e NVCC_DRAM_2P5), NVCC_PLL, VDD_SNVS_CAP, VDD_USB_CAP

VDD_SOC_IN 1.4 V

VDD_HIGH_IN e NVCC 3.3 V: un 100 nF per dominio (CSI, ENET, GPIO, LCD, NAND, SD1, UART)

NVCC_DRAM 1.35 V (lato SoC)

VDD_SNVS_IN

Y1
24MHz

Carico 24 MHz (CL da verificare sul quarzo YXC)

Y2
32.768kHz

Carico RTC: circa 2xCL meno parassite (datasheet, tabella 7)

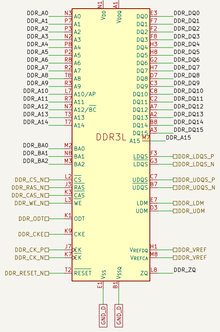
Boot: BOOT_MODE=10. BOOT_CFG1[4] a [6] DNP fische' lo strap QSPI non e' confermato sulla fuse map: JTAG_MOD a GND con 4.7k (tabella 3)

SW1
RESET

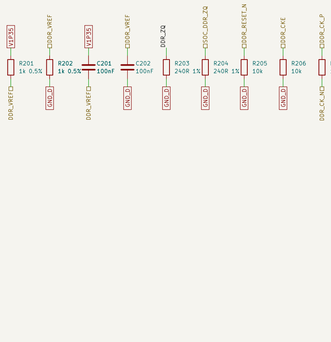
SW2
ON/OFF

U2 Micron MT41K512M16HA-107:A. 8 Gb x16 = 1 GB. FBGA-96 9x14 passo 0.8. a 1.35 V. Un solo chip punto-punto: niente VTT. Simbolo da MT41K256M16HA di KiCad con A15 su M7 (ball map JEDEC x16).

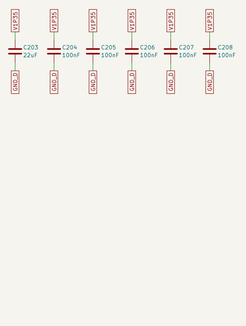
VREF a partitore 1k 0.5% con 100 nF su ogni resistenza (datasheet 6ULL, tabella 3), unica per DRAM_VREF, VREFCA, VREFDQ. ZQ 240R 1% su SoC e DRAM. RESET# e CKE con 10k a GND. CK con 100R differenziale vicino alla DRAM (da confermare in simulazione).



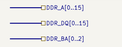
VREF, ZQ, pull-down



VDD / VDDQ 1.35 V







Sheet: /DDR3L/		
File: ddr3l.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/10

VIN_SYS 12 V dal BUS_POWER -> TPS56637 5 V (U22) -> TPS2121 U17 (IN1 bus con priorità, IN2 VBUS USB-C 1) -> 5V_SYS -> buck 3.3 V (U23). 1.35 V DDR (U26). 1.4 V VDD_SOC_IN (U27).

Sequenza 3.3 -> 1.35 -> 1.4 V a RC sugli EN del TLV62569 (soglia EN max 1.2 V): 100k/470nF circa 21 ms, 100k/1uF circa 45 ms. POR_B dal TPS3839K33 (2.93 V, push-pull) con 1k in serie, così SW1 può cortocircuitare POR_B.

Niente pila di backup: VDD_SNVS_IN e V_BCKP del GNSS direttamente al 3.3 V. RTC e dati GNSS si perdono a ogni spegnimento. Passivi 0603 minimo; 47 uF e 10 uF/25 V in 1206, 22 uF e 10 uF in 0805.

U22
TPS56637RPAR
BUCK_EN
BUCK_FB
GND_D
BUCK_PG
EN
FB
AGND
PG
NC
MODE
PGND
VIN
BOOT
SW
VIN_SYS
BUCK_BST
BUCK_SW
L1
3.3uH/8A
5V_BUCK
TPS56637: EN 169k/36.1k, FB 73.2k/10k = 4.99 V, FF 20k+100pF, MODE aperto = FCCM, 2x47 uF ai posto di 4x22 uF
VIN_SYS
C301
10uF/25V
C302
10uF/25V
C303
100nF/50V
R301
169k 1%
BUCK_EN
R302
36.1k 1%
R303
100k
BUCK_PG
C304
100nF
BUCK_BST
BUCK_SW
R304
73.2k 1%
BUCK_FB
R305
10k 1%
R306
20k
BUCK_FF
C305
100pF
C306
47uF/10V
C307
47uF/10V
U17
TPS2121RUXR
5V_BUCK
MUX_PR1
MUX_DV1
VBUS1
MUX_DV2
GND_D
MUX_SS
MUX_ILM
IN1
PR1
DV1
IN2
DV2
CP2
SS
ILIM
OUT
ST
GND
5V_SYS
CIPWR_SRC_ST
TPS2121 (come RK3568): PR1 4.49 V, OV 5.98 V, ILM 22.1k = 4-5 A. SS 100 nF
R307
32.4k 1%
R308
10k 1%
R309
46.4k 1%
R310
10k 1%
R311
46.4k 1%
R312
10k 1%
R313
22.1k 1%
C308
100nF
C309
1uF/10V
C310
1uF/10V
C311
47uF/10V
U23
TPS563201DDCR
GND_D
VBST
SW
VIN
GND
FB
EN
VFB
U23_BST
U23_EN
U23_FB
U23_SW
L2
2.2uH/9A
3V3
U23 3.3 V / 3 A: FB 33.2k/10k = 3.32 V, primo della sequenza
C312
10uF/10V
R314
100k
U23_EN
C313
100nF
U23_SW
U23_BST
R315
33.2k 1%
U23_FB
R316
10k 1%
C314
47uF/10V
U26
TLV62569DBVR
U26_EN
U26_FB
U26_SW
L3
2.2uH/3A
U26_SW
V1P35
U26 1.35 V DDR: FB 150k/120k (0.6 V x 2.25), EN 100k/470nF da 3V3
C315
10uF/10V
R317
100k
U26_EN
C316
470nF
R318
150k 1%
U26_FB
R319
120k 1%
C317
22uF/10V
U27
TLV62569DBVR
U27_EN
U27_FB
U27_SW
L4
2.2uH/3A
U27_SW
V1P4
U27 1.4 V VDD_SOC_IN: FB 133k/100k = 1.398 V (min 1.375 V, 125 mV sopra VDD_ARM_CAP). EN 100k/1uF da 3V3
C318
10uF/10V
R320
100k
U27_EN
C319
1uF
R321
133k 1%
U27_FB
R322
100k 1%
C320
22uF/10V
U28
TPS3839K33DBZR
RESET
POR_DRV
POR_B
#FLG350
PWR_FLAG
#FLG351
PWR_FLAG
#FLG352
PWR_FLAG
#FLG353
PWR_FLAG
#FLG354
PWR_FLAG
#FLG355
PWR_FLAG
#FLG356
PWR_FLAG
#FLG357
PWR_FLAG
R323
1k
C321
100nF
POR_B1

Sheet: /Alimentazione/
File: alimentazione.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 4/10

J11 BUS_POWER 30 pin (Hirose FH12-30S) su ECSPI2. J12 MAIN 28 pin REV5.1 (XUNPU FPC-05F-28PH20, pin 28 = HIT#) su ECSPI1. Mappa del connettori identica a SchedaA133/RK3568. Rail BUS_* passanti, non collegati ai rail della scheda.

Serie 33R solo sul clock SPI. Pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C. VREF_ENABLE 10k e VBIAS_ENABLE 100k a GND. CS e FAULT con i pull-up interni del 6ULL (pad control nel device tree). SPI_ID a GND diretti. ESD TPD4E05U06DQA sulle linee logiche.

J11
BUS_POWER_MIKILAB

J12
BUS_MIKILAB_MAIN

Serie SPI 33R, solo clock

Livelli di riposo

U29
TPD4E05U06DQAR

U30
TPD4E05U06DQAR

U31
TPD4E05U06DQAR

U32
TPD4E05U06DQAR

U33
TPD4E05U06DQAR

U34
TPD4E05U06DQAR

Disaccoppiamento connettori

#FLG450
PWR_FLAG

#FLG451
PWR_FLAG

#FLG452
PWR_FLAG

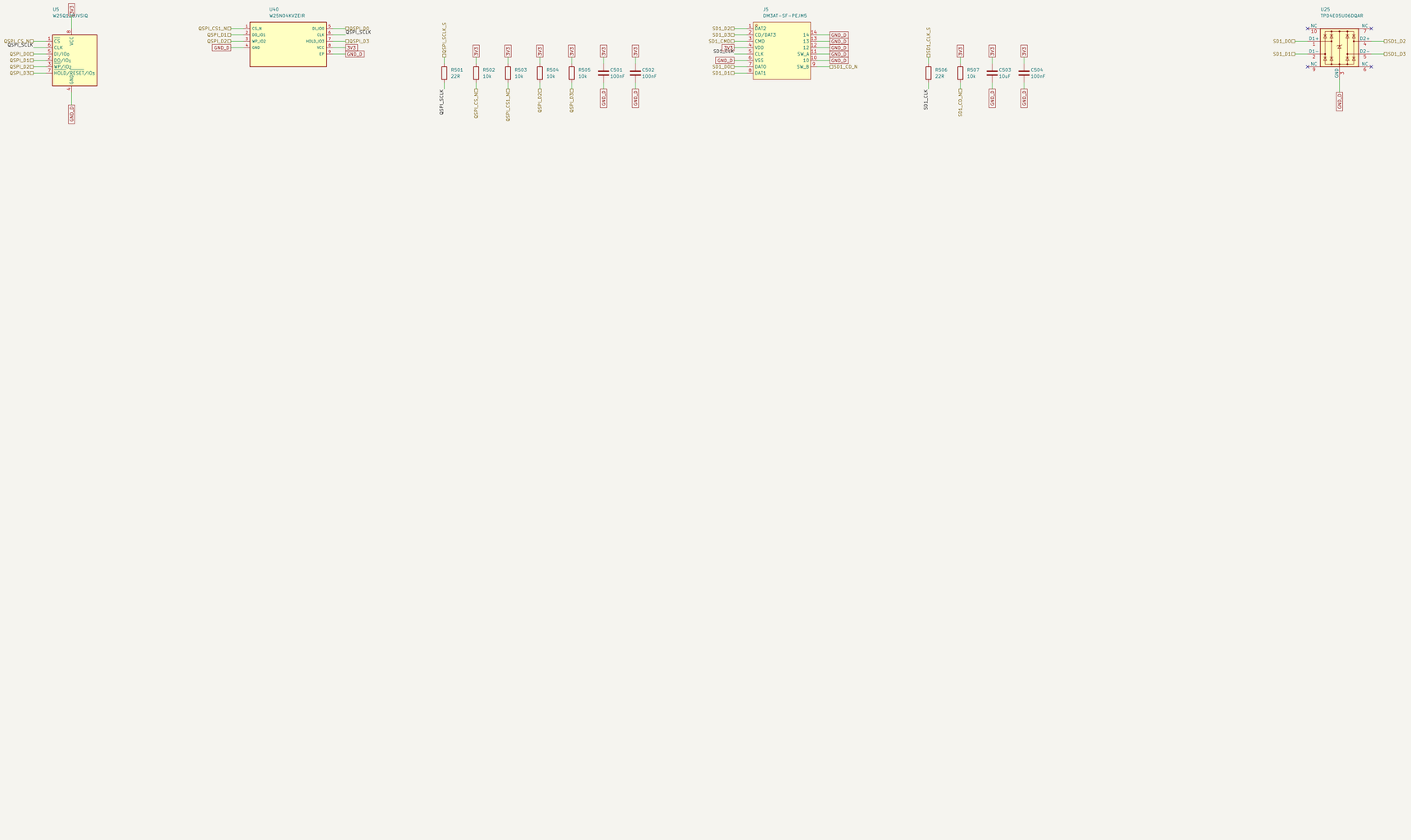
Sheet:	/Bus MIKILAB/
File:	bus-mikilab.kicad_sch
Title:	
Size:	A1
Date:	
Rev:	
KiCad E.D.A.	10.0.7
Id:	5/10

U5 W25Q128JV 16 MB su QSPI1 porta A, SS0: boot. U40 W25N04KV 512 MB SPI NAND SLC su SS1 (NAND_DATA07): rootfs UBIFS. Bus QSPI in comune. Il ROM del 6ULL non avvia da SPI NAND: per questo restano due chip.

J5 microSD DM3AT su uSDHC1 (sul BOTTOM). 3.3 V. CD su UART1_RTS_B (USDHC1_CD_B). U25 protegge DAT0–3: CMD e CLK non hanno ESD (un solo TPD4E in BOM, 4 linee).

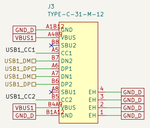
QSPI: 22R su SCLK, pull-up 10k su CS0/CS1/IO2/IO3

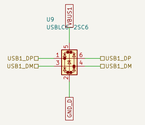
microSD: 22R su CLK, pull-up 10k su CD; CMD e DAT con i pull-up interni del 6ULL



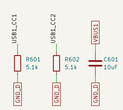
J3 USB-C 1: USB_OTG1, device/download seriale e alimentazione (VBUS1 -> IN2 del TPS2121). Sink: 5,1k su CC1/CC2. J8 USB-C 2 host: USB_OTG2 diretto, VBUS dal TPS2553 (~1 A). Rp 56k. Le due USB-C sono affiancate sullo stesso bordo.

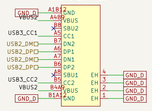
USB_OTG2_VBUS del 6ULL da 5V_SYS. Niente hub: J8 e' l'unica porta host.

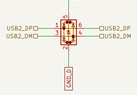




CC sink J3

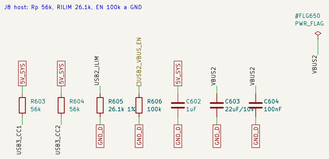








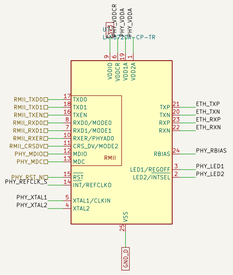
J8 host: Rp 56k. RILIM 26.1k. EN 100k a GND

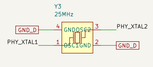


Sheet: /USB/		
File: usb.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/10

U7 LAN8720A su ENET1 RMII. Y3 25 MHz montato: il PHY genera i 50 MHz (nINTSEL basso) su REFCLK0 verso ENET1_TX_CLK (REF_CLK in ingresso, SION). nINT quindi non disponibile. Strap MODE=111 (10k su RXD0/RXD1/CRS_DV). PHYADD=0.

J1 HR911105A: magnetici e LED integrati. TX/RX con 49.9R a VDDA. CT a VDDA con 22 nF. Polarita' del LED dal disegno Hanrun: da verificare (solo indicazione, non blocca il link). U6 24AA02E48 su I2C1 con il MAC EUI-48.

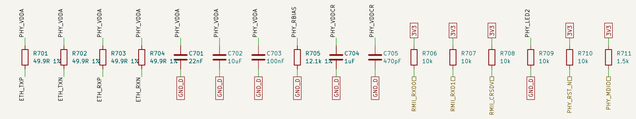


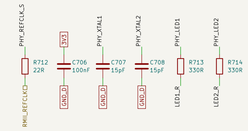




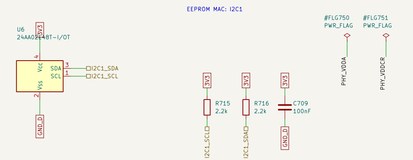


PHY: terminazioni, RBIAS 12.1k, VDDCR 1uF+470pF, straps, REFCLK 22R





EEPROM MAC: I2C1

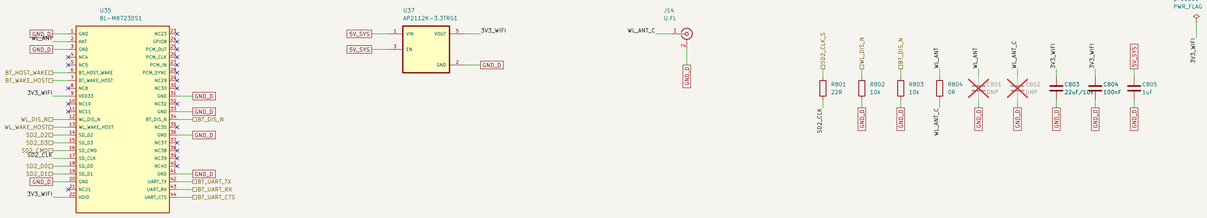


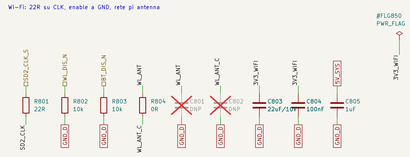
Sheet: /Rete/
File: rete.kicad_sch
Title:
Size: A1 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 8/10

U35 LB-LINK BL-M8723DS1 (RTL8723DS, 12x12 mm, datasheet V1.0.1.3): Wi-Fi su uSDHC2 (pad NAND), BT su UART8 (pad ENET2). VDD33 e VDIO da U37 (3.3 V dedicati ai picchi TX). WL_DIS# e BT_DIS# senza pull-up nel modulo: 10k a GND, li alza Linux.

SDIO con i pull-up interni del 6ULL. Antenna su U.FL J14 (scelta del placement: la BOM lasciava u.FL o PCB) con rete pi OR + 2 DNP da tarare. UART_CTS del modulo da UART8_RTS. Il modulo non ha RTS (pin 41 = GND): H5 a 3 fili. PCM non usata: PCM_OUT libero (strap da non alzare).

Wi-Fi: 22R su CLK, enable a GND, rete pi antenna

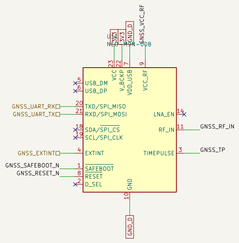




Sheet: /Radio/		
File: radio.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 9/10

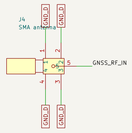
U11 u-blox NEO-M9N-00B su UART3. TIMEPULSE -> 33R -> GNSS_PPS su GPT1_CAPTURE1 (UART2_TX) e buffer U20 -> 33R -> header J7. EXTINT IN da J7 -> ESD + 100R -> buffer U19 -> GNSS_EXTINT su NEO e GPT1_CAPTURE2 (UART2_RX): evento marcato da entrambi.

J4 SMA antenna, bias 3.3 V attiva: VCC_RF -> 10R -> 27 nH (0603) -> RF_IN. 10 nF. Linea 50 ohm corta. V_BCKP al 3.3 V, niente pila: partenza a freddo a ogni accensione. J7: PPS OUT, EXTINT IN, GND (come scheda Tibbo).



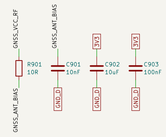


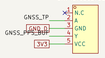


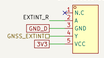




Bias antenna e alimentazione











Timing: serie e buffer



